

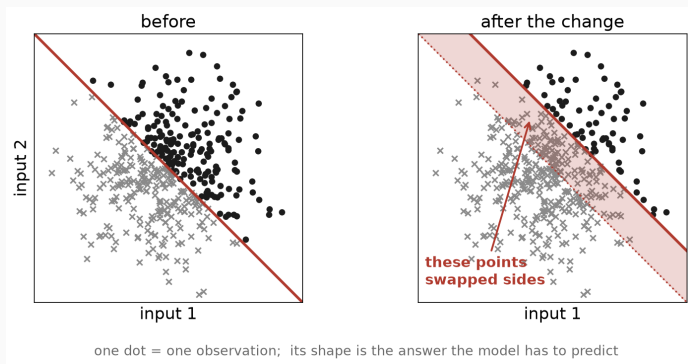
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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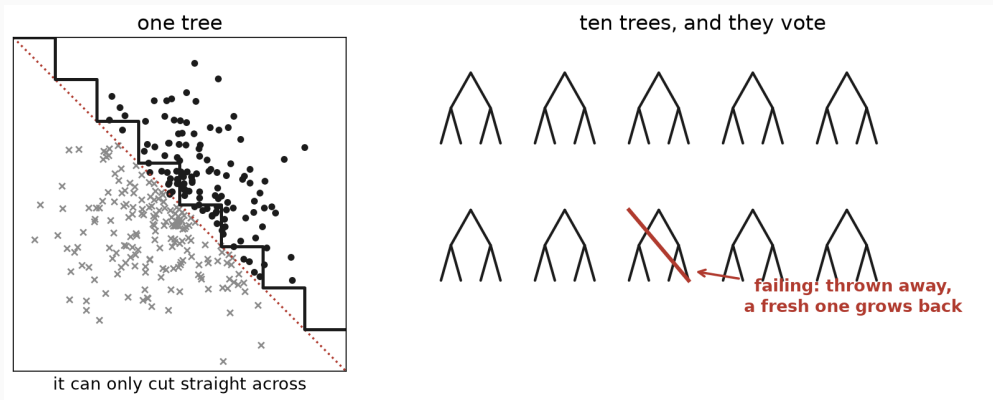
Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Defence one: a model that repairs itself

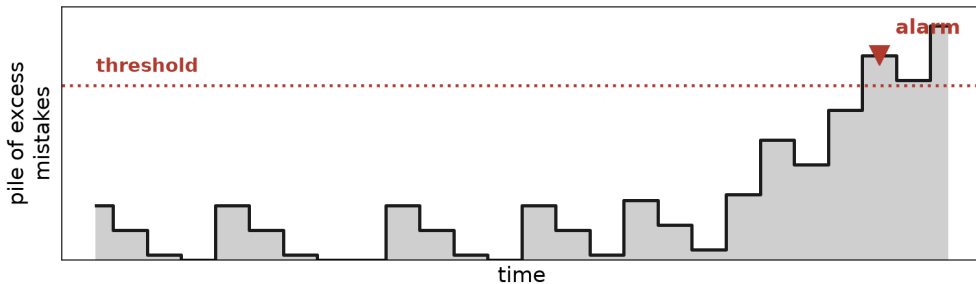


This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

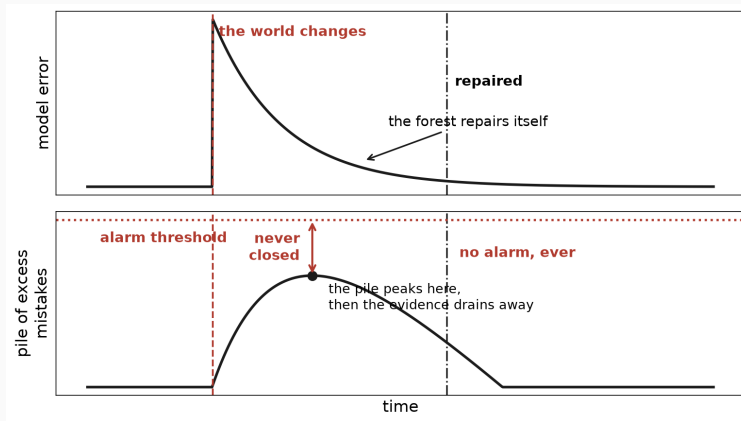
the model answers, one observation at a time (x = wrong)

x o o o x o o o o x o o o x o o x x o x x o x



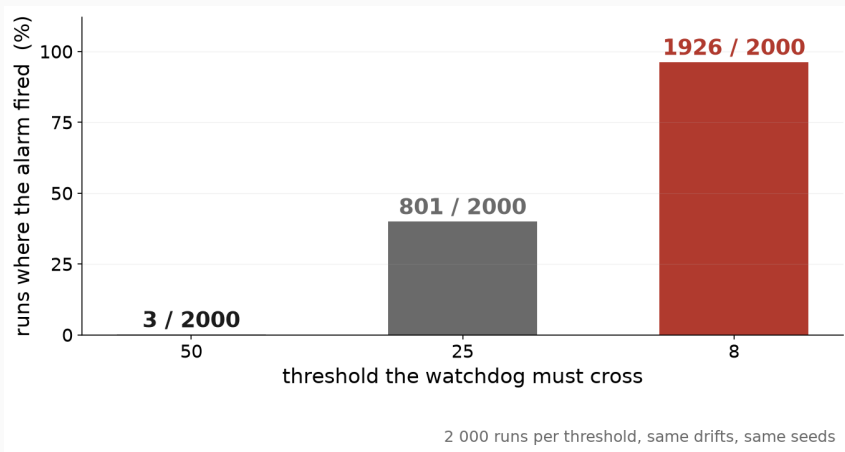
Two communities, two tools, never measured against each other.

Put them together, and they cancel out



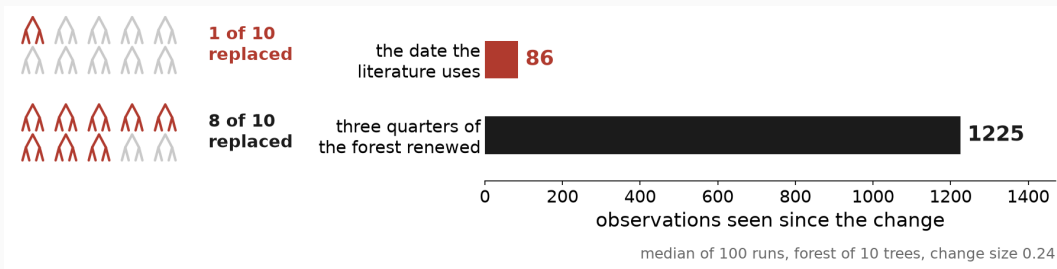
The repair has a date. The alarm never gets one. (schematic)

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

Our contribution: when is the forest actually repaired?



The accepted date fires at the *first* tree. It times the start of a repair, not its end.

Where we stand

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.